Kuanming Yao, Ph.D.

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Current Position

- **Assistant Professor | The Hong Kong University of Science and Technology | ISD & MAT & CKSRI**
- **Associate Director | Center for Smart Manufacturing**

Apr. 2026 – Now, Clear Water Bay, Hong Kong SAR, China

Academic Experience

- **Postdoc | University of California, Los Angeles | ECE** Apr. 2024 – Mar. 2026, Los Angeles
  Research interests: minimally invasive bioelectronics for organ health management by multi-modal bio-signal monitoring and early diagnosis of acute diseases.
- **Postdoc | City University of Hong Kong | BME** Feb. 2023 – Mar. 2024, Hong Kong
  Advisor: Prof. Xinge Yu (Member, HK Young Academy of Sciences)
  Research interests: bioelectronics for human-machine interfaces and cardiovascular healthcare, including sensory enhancement, soft & breathable 3D integrated electronics, and clinical-grade devices.

Education

- **Ph.D. | City University of Hong Kong | BME** 2019 – 2023, Hong Kong
  Supervisors: Prof. Xinge Yu & Prof. Lidai Wang
  Thesis: Thin, Soft, Miniaturized Instrumentation for Biomedical Sensing and Stimulation
- **B.Eng. | Zhejiang University | COEE** 2015 – 2019, Hangzhou

Academic Services

- **Journal Reviewer:** *Nature*, *Nat. Mach. Intell.*, *Nat. Sens.*, *Nat. Comm.*, *Sci. Adv.*, *Device*, *IEEE Trans. on Biomed. Eng.*, *npj Flex. Electron.*, *Adv. Func. Mater.*, *Adv. Sci.*, *Natl. Sci. Rev*, *Biosense. Bioeletron. & Small Methods*, *Research*, *Mater. Today Physics*, *Chem. Eng. J.*, *Soft Science*, *Adv. Comp. and Hyb. Mater.*, *IEEE OJ. Nano.*, *Biomed. Microdev.*, *Int. J. Hum-Comput. Int.*
- **Grant Reviewer:** The Hong Kong Research Institute of Textiles and Apparel (HKRITA)
- **Conference Reviewer:** *IEEE International Conference on Robotics and Biomimetics (ROBIO 2024)*
- **Editorial board member:** *BMEMat, Discover Electronics, Biosensors* (Guest Editor).

Qualifications

- Member, Materials Research Society (MRS)
- Member, International Society of Bionic Engineering (ISBE)
- Member, Chinese Society of Biomedical Engineering (CSBME)
- Member, Chinese Institute of Electronics (CIE)

Publications (selected)

† (co-1st), * (corresponding)

1. **K. Yao†**, J. Zhou†, …, W. J. Li*, X. Yu*, Encoding of tactile information in hand via skin-integrated wireless haptic interface, ***Nature Machine Intelligence***, 4, 893-903 (2022). **(Cover, Oct. Issue)**
2. **K. Yao†,** Q. Zhuang†, …, Z. Zheng*, X. Yu*, A fully integrated haptic textile. ***Science Advances,*** 10, eadq9575 (2024).
3. **K. Yao*,** Q. Zhuang*, Self-decoupling three-axis forces in a simple sensor, ***Nature Machine Intelligence,*** 6, 1431-1432 (2024).
4. Q. Zhuang†, **K. Yao†**, …, X. Yu*, Z. Zheng*, Permeable, three-dimensional integrated electronic skins with stretchable hybrid liquid metal solders, ***Nature Electronics,*** 7, 598-609 (2024).
5. Q. Zhuang†, **K. Yao†**, …, X. Yu*, Z. Zheng*, An ICU-grade breathable cardiac electronic skin for health, diagnostics, intraoperative and postoperative monitoring, ***Science Advances,*** 11, eadu3146 (2025).
6. Y. Huang†, **K. Yao†**, …, X. Yu*, Bioelectronics for electrical stimulation: materials, devices and biomedical applications. ***Chemical Society Reviews***, 53, 8632-8712 (2024).

7. Y. Gao†, **K. Yao†**, …, X. Yu*, Advances in Materials for Haptic Skin Electronics. ***Matter*****,** 7, 2826–2845 (2024).
8. D. Li†, J. Zhou†, **K. Yao†**, …, Y. Dai*, Z. Xie*, X. Yu*, Touch IoT enabled by wireless self-sensing and haptic-reproducing electronic skin. ***Science Advances*****,** 8, eade2450 (2022).
9. J. Zhou†, J. Li†, H. Jia†, **K. Yao†**, …, X. Yu*, Mormyroidea-inspired E-skin for active non-contact three-dimensional sensation, ***Nature Communications*****,** 15, 9875 (2024).
10. **K. Yao†**, Y. Liu†, …, Z. Xie*, L. Wang*, X. Yu*, Mechanics designs-performance relationships in epidermal triboelectric nanogenerators. ***Nano Energy***. 76, 105017 (2020).
11. **K. Yao**, …, L. Wang*, X. Yu*, Recent Advances in Materials, Designs and Applications of Skin Electronics. ***IEEE Open Journal of Nanotechnology***, 4, 55-70, (2022).
12. **K. Yao**, …, Z. Xie*, X. Yu*, Stretchable self-powered epidermal electronics from piezoelectric rubber for tactile sensing. ***Acta Physica Sinica*****,** 69, 178701 (2020).
13. T. H. Wong†, Y. Liu†, J. Li†, **K. Yao†**, …, Z. Xie*, X. Yu*, Triboelectric Nanogenerator Tattoos Enabled by Epidermal Electronic Technologies. ***Advanced Functional Materials*****,** 2111269 (2021).
14. M. Wu†, **K. Yao†**, …, E. Song*, J. Yu*, X. Yu*, Self-Powered Skin Electronics for Energy Harvesting and Healthcare Monitoring. ***Materials Today Energy*****.** 21, 100786 (2021).
15. Y. Huang†, **K. Yao†**, …, X. Yu*, Recent advances in multi-mode haptic feedback technologies towards wearable interfaces. ***Materials Today Physics*****,** 22, 100602 (2021).

Others

16. H. Zhang, …, **K. Yao**, Y. Liu, X. Yu*, ***Science Advances*** **12**, eaeh1421 (2026).
17. J. Zhu, …, **K. Yao**, …, S. Emaminejad*, ***Science Translational Medicine*** **18**, eadr5493 (2026).
18. Y. Huang, …, **K. Yao**, …, X. Yu*, ***Science Advances*** **12**, eaed7673 (2026).
19. Y. Wang, …, **K. Yao**, …, Y. Xu*, L. Chang*, ***Cell*** 189, 1086-1107.e32 (2026).
20. X. Huang, …, **K. Yao**, …, X. Yu*, ***Nature Communications*** 16, 5782 (2025)
21. D. Yin, …, **K. Yao**, …, X. Yu*, C. Yu*, L. Chang*. M. Li*, ***Nature***, 642, 1051-1061, (2025)
22. C. K. Yiu, …, **K. Yao**, …, X. Yu*, ***Science Advances***, 11, eadt6041 (2025)
23. G. Xu, …, **K. Yao**, …, X. Yu*, ***Science Advances***, 11, eadt0318 (2025)
24. B. Zhang, …, **K. Yao**, …, X. Yu*, ***Nature***. 628, 84-92 (2024).
25. M. Wu, **K. Yao**, X. Yu*, ***Nature Electronics*** 7, 731–732 (2024).
26. D. Li, …, **K. Yao**, …, L. Chang*, Z. Xie*, X. Yu*, ***Science Advances*****,** 10, eadk6301 (2024)
27. K. Nan*, …, **K. Yao**, …, X. Yu*, G. Traverso*, ***Nature Communications***, 15, 6749 (2024).
28. J. Li, …, **K. Yao**, …, X. Yu*, G. Traverso*, ***Nature Communications***, 15, 7800 (2024).
29. Y. Huang, …, **K. Yao**, …, Z. Xie*, K. Zhu*, X. Yu*, ***Nature Electronics*****,** 6, 1020–1031 (2023).
30. Q. Zhuang, **K. Yao**, …, X. Yu*, Z. Zheng*, ***Science Advances***, 9, eadg8602. (2023).
31. J. Li, …, **K. Yao,** …, D. Lei*, X. Yu*, ***Science Advances*****,** 9, eadg1837. (2023).
32. Y. Liu, …, **K. Yao**, ..., Y. Li*, X. Yu*, ***Nature Communications*****,** 14, 2297. (2023)
33. J. Li, …, **K. Yao**, …, X. Ma*, X. Yu*, ***Nature Communications*****,** 14, 5009. (2023)
34. Y. Liu, …, **K. Yao**, …, Z. Xie*, X. Yu*, ***Science Advances*****,** 8, eabl6700 (2022).
35. E. Song, …, **K. Yao**, …, J. A. Rogers*, ***Nature Biomedical Engineering*****,** 5, 759–771 (2021).
36. H. Wang†, J. Wang†, **K. Yao**, …, Z. L. Wang*, Y. Zi*, ***Science Advances*****,** 7, eabi6751 (2021).

Conferences & Talks

- **K. Yao** (Invited speaker, session chair), Bioelectronic Interfacial Systems for Sensory and Health. 8th International Conference of Bionic Engineering (***ICBE 2026***), Beijing, China. 2026.10.18
- **K. Yao** (Invited speaker), Bioelectronic Interfacial Systems for Sensory and Health. 7th International Conference on Flexible Electronics (***ICFE 2026***), Jiaxing, China. 2026.8.6
- **K. Yao** (Invited speaker), Bioelectronic Interfacial Systems for Sensory and Health. Institute of Fiber

Materials and Devices, **Fudan University**, Shanghai, China. 2026.8.4

- **K. Yao**, …, X. Yu* (Oral Speaker), Encoding tactile information in hand via skin-integrated wireless haptic interface, ***MRS fall meeting 2022***, Boston, US. 2022.11.29
- M. Wu, **K. Yao,** J. Yu*, X. Yu* (Poster), Recording of brain activity using multichannel, biocompatible organic electrochemical transistors in vivo, 2023 IEEE 18th International Conference on Nano/Micro Engineered and Molecular Systems **(*NEMS 2023*)**, Jeju.
- **K. Yao**, X. Yu* (Poster), Encoding tactile information in hand via skin-integrated wireless haptic interface, The 5th International Conference on Flexible Electronics **(*ICFE 2023*)**, Hangzhou.
- **K. Yao** (online live talk), Fully integrated wearable electrotactile interfaces and personalized virtual touching experiences, **ICANX Youth Talks Vol. 80, 2024**
- **K. Yao** (invited online talk), Integrated electronic interfaces for sensory and bio-signals, International Institute for intelligent Nanorobots and Nanosystems, **Fudan University**
- **K. Yao,** … (Poster), Cascaded Dehydrogenase Biosensors for Tracking Carbohydrate Metabolism, ***ACS Fall 2025***, Washington D.C.

Patents

- Miniaturized On-Skin Electrotactile Apparatus, X. Yu, **K. Yao** & J. Zhou, US 11954254B2 (US Patent)
- Haptic Apparatus for Stimulating Users' Skin, X. Yu, Z. Zheng, **K. Yao,** Q. Zhuang, Q. Zhang & J. Zhou, Filed, 18/974,954. (US Patent)
- Air Ventilation Optimizing Device Based on Vortex Ring Mechanism, J. Wei, T. Jin, **K. Yao**, et al. (201810494952.1) (patent for invention, China)

Awards & Highlights

- 1st Place Winner Poster Award of Clinical Translational Research category, Department of Medicine's Research Day, UCLA Health — 2025
- **Outstanding Academic Performance Awards** (CityU) × 3
- Research Tuition Scholarships (CityU) × 3 — 2020-2023
- **HKMHDIA Student Research Award** (Hong Kong Medical and Healthcare Device Industries Association) — 2021-2022
- **Cover article** of Vol. 4 Issue 10, Oct 2022, ***Nature Machine Intelligence*** — 2022.10
- **Cover article** of Vol. 53, Issue 17, Sept 2024, ***Chemical Society Reviews*** — 2024.9
- The Second-Class Prize of 11th National University Student Social Practice and Science Contest on Energy Saving & Emission Reduction — 2018.8
- The Second-Class Prize of Danfoss Cup China HVACR Innovation Award — 2018.8
- Scholarship for Excellence in Research and Innovation (ZJU) — 2017-2018